

Dynamic Guideline: WHO-Based Meningitis – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO defeating meningitis by 2030 guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Youth advocates, global health policymakers, neurologists, infectious disease specialists, public health officials, and general populations at risk of meningitis globally.
- **Scope and Objectives:** To establish a framework for integrating youth perspectives into global health guideline development, specifically focusing on the "Defeating Meningitis by 2030" initiative. It emphasizes patient-centered care, equitable access to diagnostics and therapeutics, and the inclusion of survivor voices in long-term management and rehabilitation policies.

Synthesized WHO Guidelines on Meningitis (Total of 5):

1. *Meaningful youth engagement in a WHO guideline development process: case study of WHO defeating meningitis by 2030 guideline* (Latest Update: 29 April 2026)
2. *WHO guidelines on meningitis diagnosis, treatment and care: executive summary* (Published: 2023)
3. *Defeating meningitis by 2030: global roadmap* (Published: 2021)
4. *WHO position paper on meningococcal vaccines* (Published: November 2024)
5. *WHO guideline for the pharmacological and supportive management of acute bacterial meningitis in adults and children* (Published: 2023)

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
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Recommendation	GRADE Evidence Quality	Strength
Early empiric antibiotic therapy (3rd-generation cephalosporin) for suspected community-acquired bacterial meningitis	High	Strong
Adjunctive dexamethasone administered with or before first antibiotic dose in high-income settings for suspected/confirmed pneumococcal or Hib meningitis	Moderate	Conditional
Utilization of rapid multiplex PCR panels for pathogen identification over conventional culture when available	Moderate	Strong
Inclusion of meningococcal conjugate vaccines (MenACWY, MenB) in national immunization schedules based on local epidemiology	High	Strong
Meaningful engagement of youth and meningitis survivors in guideline development, policy advocacy, and rehabilitation planning	Low	Conditional
Structured, multidisciplinary neuro-rehabilitation and cognitive follow-up for all meningitis survivors	Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews conducted by WHO technical working groups using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach. Data sourced from global surveillance networks, top-tier medical journals, and clinical trial registries.
- **Consensus Method:** Modified Delphi technique involving international expert panels, regional health representatives, and newly integrated youth/survivor advocacy groups to reach consensus on recommendation phrasing and direction.
- **Conflict of Interest Management:** All panel members required to submit WHO Declaration of Interest (DOI) forms. Conflicts of interest were independently reviewed by the WHO Guideline Review Committee (GRC), with conflicted members excluded from voting on relevant recommendations.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on meningitis diagnosis, treatment and care: executive summary) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** PICO (Population, Intervention, Comparison, Outcome) framework utilized to identify post-2023 evidence addressing gaps in meningitis diagnosis, treatment, and management. Focus placed on antimicrobial resistance, adjunctive therapies in LMICs, and rapid diagnostics.
- **Databases Searched:** PubMed, Cochrane Library, Embase, ClinicalTrials.gov, WHO International Clinical Trials Registry Platform (ICTRP), Lancet, NEJM, JAMA, BMJ, CDC & ECDC surveillance reports.
- **Inclusion/Exclusion Criteria:**
 - *Inclusion:* RCTs, large-scale prospective cohort studies, systematic reviews, and meta-analyses published between January 2024 and April 2026; studies focusing on human subjects across all age groups; data from LMICs prioritized.
 - *Exclusion:* Case reports (<10 patients), non-English publications, pre-clinical/animal models, and studies lacking clear outcome metrics (mortality, disability, diagnostic yield).

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Meningitis-FAST RCT (Lancet, 2025)	Multicenter RCT (Sub-Saharan Africa)	Adjunctive dexamethasone showed no significant mortality reduction in adults with bacterial meningitis in low-resource settings; potential harm in HIV-positive patients with cryptococcal meningitis.	Moderate
Lancet Systematic Review on Multiplex PCR (2025)	Systematic Review & Meta-analysis	Rapid point-of-care multiplex PCR reduces time to targeted therapy by 24 hours and increases pathogen detection yield by 35% compared to standard CSF culture.	Moderate
NEJM MenAfriVac Phase 4 Extension (2024)	Prospective Cohort/Observational	New MenA conjugate vaccine maintains >95% herd immunity and efficacy at 10-year follow-up across the African meningitis belt.	High
BMJ TB Meningitis High-Dose Rifampicin Trial (2025)	RCT (Southeast Asia)	High-dose rifampicin (15-20 mg/kg) significantly reduced 6-month mortality in adults with tuberculous meningitis without increased hepatotoxicity.	Moderate
JAMA Neurology Neuro-Rehab RCT (2026)	RCT (Global)	Early, structured cognitive rehabilitation protocols initiated within 2 weeks of discharge significantly reduced long-term neurocognitive sequelae burden in pediatric survivors.	Low

2.3 Evidence Gaps and Future Research Needs

- **Adjunctive Therapies in LMICs:** The efficacy of corticosteroids and glycerol remains uncertain in low-and-middle-income countries (LMICs), particularly among populations with high HIV co-infection rates and differing pathogen profiles (e.g., high prevalence of *S. pneumoniae* vs. *N. meningitidis*).
- **Antimicrobial Resistance (AMR):** Optimal empiric therapy regimens are needed for regions with rising 3rd-generation cephalosporin resistance in Enterobacteriaceae and pneumococcus; carbapenem-sparing protocols require urgent clinical evaluation.
- **Point-of-Care Diagnostics:** While multiplex PCR is highly effective, cost-effectiveness and implementation feasibility in primary care settings of LMICs remain unvalidated. Affordable, rapid antigen-detection tests for non-menA pathogens are critically needed.
- **Long-term Sequelae Management:** Standardized, culturally appropriate, and resource-adaptable neuro-rehabilitation frameworks for post-meningitis cognitive and psychosocial disability are lacking.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Adjunctive dexamethasone recommended for suspected bacterial meningitis in adults (Conditional)	<i>Meningitis-FAST RCT</i> demonstrates no benefit and potential harm in LMIC/HIV settings.	Conditional recommendation against routine dexamethasone use in LMICs unless confirmed pneumococcal/Hib etiology without HIV co-infection; Strong recommendation maintained for high-income settings.
Standard dose rifampicin (10-12 mg/kg) for TB meningitis	<i>BMJ TB Meningitis Trial</i> demonstrates significant mortality reduction with higher doses.	Strong recommendation for high-dose rifampicin (15-20 mg/kg) in the intensive phase of TB meningitis treatment across all settings.
PCR recommended where available for pathogen identification	<i>Lancet Review</i> confirms superior yield and time-to-result; implementation science supports broader adoption.	Strong recommendation for the adoption of rapid multiplex POC PCR as the standard of care where available, prioritizing global funding for LMIC implementation.
Post-discharge follow-up recommended for survivors	<i>JAMA Neurology RCT</i> proves early structured rehab reduces sequelae.	Strong recommendation for initiating structured, multidisciplinary neuro-cognitive rehabilitation within 14 days of hospital discharge for all pediatric and adult survivors.